

OBINexus Truth: Formal Proof Framework

Eziokwu - Mathematical Foundation for AI Accountability

OBINexus Systems

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Abstract

This document establishes the formal mathematical framework for OBINexus Truth's proof verification system, implementing the Declarative-Constructive hybrid logic through rigorous set theory and procedural definitions. The framework provides structured mechanisms for handling claimed proofs within AI accountability infrastructure while maintaining cultural sensitivity and human-centered design principles.

Contents

1	Foundation Set Definitions	2
2	Declarative Proof Logic (7-Method Model)	2
3	Constructive Proof Overlay	3
4	Formal Procedure for Claimed Proof Handling	3
5	Proof Space Topology	4
6	Implementation Architecture	4
7	Governance Integration	4
8	Conclusion	4

1 Foundation Set Definitions

Definition 1.1 (Universal Evidence Space). *Let $\mathcal{U}$ be the universal evidence space containing all possible evidence artifacts:*

$$\mathcal{U} = \{e : e \text{ is a digitally representable evidence artifact}\}$$

Definition 1.2 (Claim Bubble). *A claim bubble $\mathcal{C}$ is defined as a 4-tuple:*

$$\mathcal{C} = \langle A, E, T, \Phi \rangle$$

where:

- A is the assertion set: $A = \{a_1, a_2, \dots, a_n\}$ where each a_i is an atomic claim
- $E \subseteq \mathcal{U}$ is the evidence set supporting the claims
- T is the temporal constraint set: $T = [t_{start}, t_{end}] \times \mathbb{N}$
- Φ is the attribution mapping: $\Phi : A \rightarrow \mathcal{P}(\text{Actors})$

Definition 1.3 (Proof Space). *The proof space $\mathcal{PS}$ for a claim bubble $\mathcal{C}$ is defined as:*

$$\mathcal{PS}(\mathcal{C}) = \{p : p \text{ is a valid proof sequence for } \mathcal{C}\}$$

where each proof $p \in \mathcal{PS}(\mathcal{C})$ must satisfy both declarative and constructive constraints.

2 Declarative Proof Logic (7-Method Model)

Definition 2.1 (Declarative Proof Sequence). *A declarative proof sequence $D = \langle d_1, d_2, \dots, d_7 \rangle$ for claim bubble $\mathcal{C}$ is defined where:*

$$d_1 : \text{Assertion} \rightarrow A \text{ (traceable claim establishment)} \tag{1}$$

$$d_2 : \text{Evidence} \rightarrow E \text{ (data and context provision)} \tag{2}$$

$$d_3 : \text{Attribution} \rightarrow \Phi \text{ (actor/system association)} \tag{3}$$

$$d_4 : \text{Cross-Verification} \rightarrow V(\mathcal{C}) \text{ (independent validation)} \tag{4}$$

$$d_5 : \text{Contradiction Check} \rightarrow \neg \exists c \in \mathcal{C}^c : c \models \neg A \tag{5}$$

$$d_6 : \text{Temporal Validation} \rightarrow T \models \text{occurrence constraints} \tag{6}$$

$$d_7 : \text{Archival Recording} \rightarrow H(\mathcal{C}, D) \text{ (cryptographic storage)} \tag{7}$$

Definition 2.2 (Cross-Verification Function). *The cross-verification function $V : \mathcal{C} \rightarrow \{0, 1\}$ is defined as:*

$$V(\mathcal{C}) = \begin{cases} 1 & \text{if } \exists E_{ind} \subseteq \mathcal{U} : E_{ind} \cap E = \emptyset \wedge E_{ind} \models A \\ 0 & \text{otherwise} \end{cases}$$

where E_{ind} represents independently sourced evidence.

3 Constructive Proof Overlay

Definition 3.1 (Constructive Proof Properties). *A constructive proof K for claim bubble $\mathcal{C}$ must satisfy:*

1. **Executable Sequence:** $\exists f : \mathcal{C} \rightarrow \mathcal{O}$ where $\mathcal{O}$ is the set of observable outcomes
2. **Fault Traceability:** $\forall \text{failure } \phi \in K, \exists \text{trace } \tau : \tau \models \text{reconstruction}(\phi)$
3. **Evidence Transformability:** $\exists g : E \times \text{Context} \rightarrow E'$ where $E' \models A$ in new context
4. **No Black Box Doctrine:** $\forall \text{result } r \in K, \exists \text{logic chain } \ell : \ell \vdash r$

Theorem 3.1 (Proof Completeness). *A proof p is complete for claim bubble $\mathcal{C}$ if and only if:*

$$p \in \mathcal{PS}(\mathcal{C}) \wedge D(p) = 1 \wedge K(p) = 1$$

where $D(p)$ indicates declarative validity and $K(p)$ indicates constructive validity.

4 Formal Procedure for Claimed Proof Handling

Procedure 4.1 (Claimed Proof Verification Protocol). *Given a claimed proof $\hat{p}$ for claim bubble $\mathcal{C}$, execute the following verification sequence:*

Input: Claimed proof $\hat{p}$, Claim bubble $\mathcal{C}$ **Output:** Verification result $v \in \{ \text{VALID}, \text{INVALID}, \text{INSUFFICIENT} \}$

1. **Claim Bubble Validation:**

$$\text{Validate}(\mathcal{C}) = \text{check}(A \neq \emptyset) \wedge \text{check}(E \subseteq \mathcal{U}) \quad (8)$$

$$\wedge \text{check}(T \text{ is well-formed}) \quad (9)$$

$$\wedge \text{check}(\Phi \text{ is total}) \quad (10)$$

2. **Evidence Integrity Check:**

$$\forall e \in E : \text{verify}(\text{hash}(e)) \wedge \text{validate}(\text{metadata}(e))$$

3. **Declarative Verification:**

$$D_{\text{valid}}(\hat{p}) = \bigwedge_{i=1}^7 d_i(\hat{p}) = \text{true}$$

4. **Constructive Verification:**

$$K_{\text{valid}}(\hat{p}) = \text{executable}(\hat{p}) \wedge \text{traceable}(\hat{p}) \wedge \text{transformable}(\hat{p}) \wedge \text{transparent}(\hat{p})$$

5. **Cultural Sensitivity Assessment:**

$$\text{Cultural}(\hat{p}) = \text{respect}(\text{context}) \wedge \text{accommodate}(\text{neurodivergence}) \wedge \text{preserve}(\text{dignity})$$

6. **Final Verification Decision:**

$$v = \begin{cases} \text{VALID} & \text{if } D_{\text{valid}}(\hat{p}) \wedge K_{\text{valid}}(\hat{p}) \wedge \text{Cultural}(\hat{p}) \\ \text{INVALID} & \text{if } \neg D_{\text{valid}}(\hat{p}) \vee \neg K_{\text{valid}}(\hat{p}) \vee \neg \text{Cultural}(\hat{p}) \\ \text{INSUFFICIENT} & \text{otherwise} \end{cases}$$

5 Proof Space Topology

Definition 5.1 (Proof Distance Metric). *For two proofs $p_1, p_2 \in \mathcal{PS}(\mathcal{C})$, define the proof distance:*

$$d(p_1, p_2) = \sum_{i=1}^{\tau} w_i \cdot |d_i(p_1) - d_i(p_2)| + \lambda \cdot |K(p_1) - K(p_2)|$$

where w_i are weights for declarative components and λ weights constructive differences.

Definition 5.2 (Proof Convergence). *A sequence of proofs $\{p_n\}$ converges to truth $\mathcal{T}$ if:*

$$\lim_{n \rightarrow \infty} d(p_n, \mathcal{T}) = 0$$

where $\mathcal{T}$ represents the ground truth for claim bubble $\mathcal{C}$.

6 Implementation Architecture

Definition 6.1 (Evidence Processing Pipeline). *The evidence processing function $\Pi : \mathcal{U} \rightarrow \mathcal{PS}$ is decomposed as:*

$$\Pi = H_7 \circ T_6 \circ C_5 \circ V_4 \circ A_3 \circ E_2 \circ I_1$$

where each Π_i corresponds to the i -th step of the declarative proof model.

Lemma 6.1 (Pipeline Composability). *If each stage Π_i preserves evidence integrity and cultural sensitivity, then the composed pipeline Π maintains these properties.*

7 Governance Integration

Definition 7.1 (Policy-Procedure Coupling). *For governance framework integration, define the coupling function:*

$$PMP : \mathcal{PS} \times \text{Policies} \rightarrow \text{Procedures}$$

such that every proof verification maps to executable governance procedures while maintaining the No Ghosting Protocol.

Theorem 7.1 (Accountability Preservation). *The OBINexus Truth framework preserves accountability if for every verified proof $p \in \mathcal{PS}(\mathcal{C})$:*

$$\exists \text{audit trail } \alpha : \alpha \models \text{complete traceability of } p$$

8 Conclusion

This formal framework establishes the mathematical foundation for OBINexus Truth's proof verification system, ensuring that AI accountability mechanisms maintain rigorous logical standards while preserving human-centered values and cultural sensitivity. The hybrid Declarative-Constructive approach provides both theoretical soundness and practical implementability for real-world AI transparency requirements.

Eziokwu - Truth as correct speech, implemented through heart-centered computational logic.